



Exploring combustion chemistry of ethyl valerate at various pressures: Pyrolysis, laminar burning velocity and kinetic modeling

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ABSTRACT

In this work, pyrolysis experiments of ethyl valerate were performed in a flow reactor over 705–1051 K at low and atmospheric pressures and in a jet-stirred reactor over 633–1013 K at near-atmospheric pressure. Products were measured with synchrotron vacuum ultraviolet photoionization mass spectrometry in the flow reactor pyrolysis and gas chromatography in the jet-stirred reactor pyrolysis. Valeric acid and ethylene were observed as the most abundant pyrolysis products in both experiments. Laminar burning velocities of ethyl valerate/air mixtures were also measured in a high-pressure constant-volume cylindrical combustion vessel at the initial temperature of 443 K and initial pressures of 1–10 atm. A kinetic model of ethyl valerate combustion incorporated with recent theoretical progress was developed to predict the new experimental data in this work, as well as the speciation data under flame conditions and laminar burning velocities at different initial temperatures and pressures in literature. Experimental observations and modeling analyses both confirm the significant role of the intramolecular elimination reaction of ethyl valerate producing valeric acid and ethylene. In particular, this reaction has exclusive significance in decomposition of ethyl valerate under pyrolysis conditions, indicating pyrolysis experiments can provide crucial constraints for its rate constant. Subsequent decomposition reactions of valeric acid at higher temperatures enrich the intermediate pool, especially radicals, and can continue producing ethylene to make its mole fraction keep growing under the investigated temperature ranges in the jet-stirred reactor pyrolysis. Under the flame propagation conditions, C₀–C₁ reactions have the highest sensitivity coefficients to the flame propagation, while ethylene- and vinyl-involved reactions also play important roles due to the abundant production of ethylene.

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1. Introduction

Valeric biofuels, as illustrated in Fig. 1, are a group of esters that contains the carbon skeleton from the valeric acid and an alkyl group (denoted as R). Remarkable progress has been achieved on improving the production of valeric biofuels from lignocellulose through levulinic acid or γ -valerolactone [1–4]. These fuels have promising combustion characteristics such as improved energy densities and appropriate polarities, which further facilitates their potential application as engine fuels [4,5]. In particular, depending on the length of alkyl group, valeric biofuels are compatible for either gasoline or diesel applications according to the different volatility and ignition properties [5]. Among these valeric

biofuels, methyl valerate and ethyl valerate are the two smallest and most important ones for spark-ignition engine applications [6]. Compared to methyl valerate, ethyl valerate (also known as ethyl pentanoate) shows distinct advantages like improved volumetric energy content and lower heating value (30.3 MJ/kg) [4,5,7,8]. Both spark-ignition engine test [6,7] and road trial run [4] found that the addition of ethyl valerate to gasoline has no significant impacts on engine performance. Thus ethyl valerate has been deemed as a promising fuel additive or alternative in spark ignition engine applications [6].

Several fundamental researches on combustion characteristics of ethyl valerate have been conducted in literature, including measurements of speciation data and global combustion parameters. Dayma et al. [9] measured the mole fraction profiles of stable species in the jet-stirred reactor (JSR) oxidation of ethyl valerate at 10 atm, 560–1160 K and equivalence ratios (ϕ) of 0.6, 1.0

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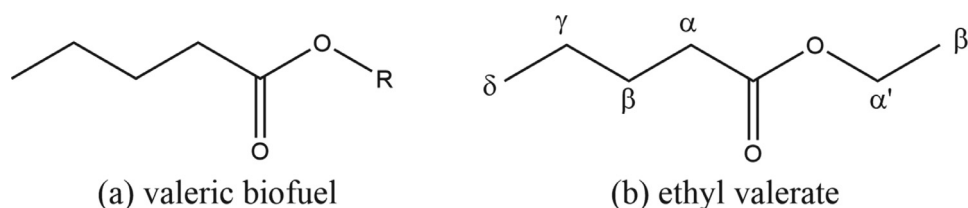


Fig. 1. A scheme of molecular structure of (a) valeric biofuels and (b) ethyl valerate. R denotes methyl, ethyl, propyl, etc.

Table 1

Experimental conditions of the flow reactor pyrolysis in this work.

P/atm	T _{max} /K	Total flow rate/SCCM*	Fuel%	Ar%	He%	Kr%
0.04	809–1051	1111.7	0.27	79.79	19.04	0.90
1	705–923	1005.8	0.30	88.19	10.52	0.99

* SCCM - standard cubic centimeters per minute.

and 2.0 using gas chromatography (GC) methods. Knyazkov et al. [10] investigated the stoichiometric laminar premixed flame of ethyl valerate at 0.026 atm using synchrotron vacuum ultraviolet photoionization mass spectrometry (SVUV-PIMS). A more recent investigation of the stoichiometric laminar premixed flame of ethyl valerate was performed by Dmitriev et al. [11] at 0.066 atm and 1 atm using GC and molecular beam mass spectrometry (MBMS), respectively. Katshiatshia et al. [12,13] studied the laminar premixed flame of ethyl valerate at 0.054 atm and various equivalence ratios using GC. Furthermore, Katshiatshia et al. [13] measured the laminar burning velocities (LBVs) of ethyl valerate at the initial pressure (P_u) of 1 atm and initial temperatures (T_u) of 318–338 K using the heat flux burner method. LBV measurements at a higher T_u (423 K) were performed by Wu et al. [14] at $P_u = 1$ atm using Bunsen flame method and Dayma et al. [9] at $P_u = 1, 3, 5$ and 10 atm using outwardly propagating spherical flame method, however discrepancies between the two measurements under same conditions has been observed up to 7 cm/s, while only one condition ($\phi = 1.0$) was reported at $P_u = 10$ atm by Dayma et al. [9]. It can be found that no investigation on the pyrolysis of ethyl valerate has been conducted, while investigations on the LBVs of ethyl valerate are also insufficient, especially under high pressures.

Regarding previous efforts in kinetic modeling work, only Dayma et al. [9] reported the development of ethyl valerate model (named as the Dayma model), which was validated against their JSR oxidation and LBV data. Further validation by Knyazkov et al. [10] in their laminar premixed flame structure study suggested that the decomposition reactions for the primary fuel radicals and their successive β -scission products in the Dayma model needed to be updated. Dmitriev et al. [11] incorporated the sub-mechanism of ethyl valerate from the Dayma model into the kinetic model of small ethyl esters developed by Sun et al. [15] with some modifications (named as the Dmitriev model) to predict their very recent flame structure data. They also suggested the need of further investigations on H-abstraction reactions of ethyl valerate and decomposition reactions of produced fuel radicals. In addition, according to investigations on ethyl esters [16,17], the intramolecular elimination of ethyl esters producing ethylene and corresponding acid has been proven to be of significant importance. Recently great efforts [15,18] have been devoted to this reaction, which have not been incorporated in kinetic models of ethyl valerate.

This work reports both experimental and kinetic modeling efforts on exploring the combustion chemistry of ethyl valerate. Pyrolysis of ethyl valerate was investigated in a flow reactor over 705–1051 K at 0.04 and 1 atm using SVUV-PIMS and in a JSR over 633–1013 K at 1.03 atm using GC. LBVs of ethyl valerate/air mixtures were also measured in a high-pressure constant-volume cylindrical combustion vessel at $T_u = 443$ K, $P_u = 1$ –10 atm and

a variety of equivalence ratios. A detailed kinetic model for the intermediate-to-high temperature chemistry of ethyl valerate was developed and validated against new experimental data in this work and previous data in literature. Modeling analysis were performed to reveal critical reactions in pyrolysis and laminar flame propagation.

2. Experimental methods

2.1. Flow reactor pyrolysis

The flow reactor pyrolysis experiments were carried out at National Synchrotron Radiation Laboratory. Detailed descriptions of the beamline and flow reactor apparatus can be found in our previous work [19–21]. In this work, the flow reactor was made of α -alumina to reduce wall catalytic effect [22–24]. The heating length of the flow reactor was extended from 224 to 294 mm, while the inner diameter was maintained at 7.0 mm. The centerline temperature profiles were measured using an S-type thermocouple following the same method in our previous work [25]. Compared with our previous apparatus [20,21], a broader plateau region was measured in the centerline temperature profiles. The uncertainties of centerline temperature were estimated to be within ± 15 K based on repeated measurements and the weak endothermicity of pyrolysis reaction [26]. The measured temperature profiles named by the maximum values are provided in the *Supplementary Materials*, which were used as input parameters in the simulation. The experimental conditions of the flow reactor pyrolysis are listed in Table 1. Ethyl valerate (98% purity from TCI (Shanghai) Development Co., Ltd, also used in the following experiments) was injected into a vaporizer and mixed with inert gasses (all 99.999% purity from Nanjing Special Gas Factory) before entering the flow reactor. The pyrolysis species were sampled by a quartz nozzle and detected by a reflectron time-of-flight mass spectrometer (RTOF-MS) combined with synchrotron VUV photoionization. Methodologies for intermediate identification and mole fraction evaluation have been reported in detail previously [27,28]. The uncertainties of measured mole fractions are estimated to be within $\pm 10\%$, $\pm 25\%$ and $\pm 25\%$ for ethyl valerate, ethylene and valeric acid, respectively.

2.2. Jet-stirred reactor pyrolysis

The JSR pyrolysis experiments were conducted at Shanghai Jiao Tong University. Detailed introduction of the quartz JSR has been reported in our previous work [29, 30], which follows the rules established by Matras and Villermaux [31]. The reactor has a spherical volume of 102 cm³ with four 0.3-mm-orifice nozzles fused inside the sphere. The mixture of fuel vapor and He (0.5%/99.5% in

mole) with a total flow rate of 1000 SCCM was fed into the reactor after passing through a short annular preheating zone. Both the reactor and the preheating zone were heated by the heating jackets and the heating temperature was adjusted by the temperature-controlling units. The temperature in the reactor was measured by a K-type thermocouple mounted in the center of the reactor, while the pressure in the reactor was maintained at 1.03 atm. Pyrolysis products were detected by an online GC (Agilent 7890B) equipped with a thermal conductivity detector (TCD) and two flame ionization detectors (FIDs). The TCD was coupled with packed columns to measure inorganic gasses (CO and CO₂), while the FIDs combined with GS-Alumina and HP-INNOWax capillary columns were used to measure hydrocarbons (mainly C₁–C₄ hydrocarbons in this work) and oxygenated compounds (ethyl valerate and valeric acid), respectively. Following methodologies in previous work [32,33], quantification of the pyrolysis species was performed by direct calibration using standard curves and indirect calibration using the effective carbon number method based on the response of FID. The uncertainties of the measured mole fractions are estimated to be $\pm 10\%$ for ethyl valerate, $\pm 15\%$ for pyrolysis products with direct calibration and $\pm 20\%$ for valeric acid and 1-butene with indirect calibration.

2.3. Laminar burning velocity

The LBV measurements were performed in a high-pressure constant-volume cylindrical combustion vessel at Shanghai Jiao Tong University, which has been introduced in detail elsewhere [20,34]. In this work, temperatures of the premixing vessel, vaporizer, pipelines and combustion vessel were all maintained at 443 K, while P_u was selected at 1, 2, 5 and 10 atm with equivalence ratios ranging from 0.7 to 1.5. Before entering the combustion vessel, the fuel vapor and synthetic air (21%O₂/79%N₂ provided by Air Liquide (Shanghai) Co., Ltd) were mixed in the premixing vessel by using partial pressure method. Then the prepared combustible mixture was ignited by the discharge of two horizontally installed tungsten electrodes in the combustion vessel. A schlieren system with a high-speed camera (Phantom V310) was used to record the flame propagation process. The camera was operated at 12,001 frames/sec with a resolution of 480 × 480 pixels, corresponding to a recorded region of 75 × 75 mm². Detailed descriptions of the data processing method can be found in the *Supplementary materials*. Briefly, the nonlinear extrapolation method developed by Kelley and Law [35] was adopted to derive the LBV and Markstein length results. The flame radius range was selected from 10 to 23 mm to eliminate the ignition and confinement effects [36–39]. The stretched burning velocities as a function of stretch rate are also illustrated in Fig. S1 in the *Supplementary materials*. According to the method reported by Moffat [40], the experimental uncertainties of measured LBVs at different pressures and equivalence ratios were evaluated, ranging from 1.06 to 1.99 cm/s. The experimental data of pyrolysis and LBV of ethyl valerate can be found in the *Supplementary Materials*.

3. Kinetic modeling

The present model consisting of 197 species and 1691 reactions was developed based on our recent models of propanal [29], butanes [20] and propanols [41]. The sub-mechanisms of ethyl valerate (EV) and valeric acid (C₄H₉COOH) were developed from the Dayma model [9] and the valeric acid model reported by Namysl et al. [42]. Updates were made based on recent theoretical calculations [18,43,44], which will be introduced in detail below. Table 2 lists the rate coefficients of some key reactions in the ethyl valerate sub-mechanism. According to previous investigations [10,15], ethylene (C₂H₄) is an important product during the combustion

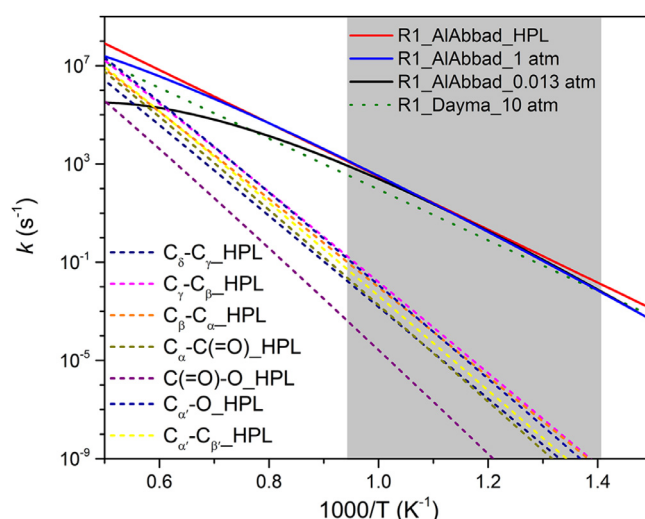


Fig. 2. Comparison of rate constants for the intramolecular elimination reaction (R1, solid and dot lines) and C–C and C–O bond dissociation reactions (dash lines) of ethyl valerate. Solid and dash lines represent the rate constants used in the present model, while dot line is the rate constant estimated by Dayma et al. [9]. Gray zone denotes the temperature range of the present flow reactor pyrolysis work.

of ethyl esters and its reactions with O were also updated based on recent theoretical progress [45]. The thermodynamic and transport data for species in the sub-mechanism of ethyl valerate were taken from the Dayma model [9]. The reaction mechanism, thermodynamic data and transport data files of the present model can be found in the *Supplementary Materials*.

3.1. Unimolecular decomposition reactions of ethyl valerate

Both the experimental observations [16,17] and theoretical calculations [47] confirmed the dominant role of intramolecular elimination in decomposition of ethyl esters. In particular, Giri et al. [16] found that ethyl propionate was almost totally consumed through this reaction in the shock tube pyrolysis at 1205 K and 2.3 atm. Great efforts on experimental measurements [16] and theoretical calculations [15,16,47,48] of intramolecular elimination of C₃–C₅ fatty acid ethyl esters have been made. Previous theoretical investigations [15,16,48] revealed that this reaction can undergo via 4- or 6-membered transition states to produce carboxylic acids and ethylene, while contributions of the pathways via 4-membered transition states are considered to be negligible due to their much higher energy barriers than those via the 6-membered transition states (up to 15 kcal/mol). Up to now, there is no literature investigation on the intramolecular elimination of ethyl valerate producing valeric acid and ethylene (R1). In the Dayma model [9], the rate constant of R1 was estimated to simulate the fuel conversion in JSR oxidation at 10 atm. Recently, theoretical investigations [18,49] have been performed to the intramolecular elimination reaction of another C₇ ethyl ester, i.e. ethyl levulinate which has a carbonyl group on γ carbon site compared to ethyl valerate. Among them, AlAbbad et al. [18] derived temperature- and pressure-dependent rate constant of the intramolecular elimination reactions of ethyl levulinate, as illustrated in Fig. 2. It is observed that their high-pressure limit (HPL) rate constant is a little faster than that in the Dayma model [9]. In the present model, the rate constant of R1 was referred to the result of ethyl levulinate reported by AlAbbad et al. [18] considering the structural similarity.

The C–C and C–O bond dissociation reactions of ethyl valerate were also incorporated in the present model. Since there is also no previous investigation for these reactions, their rate constants were

Table 2A list of selected reactions in the ethyl valerate sub-mechanism, units are s^{-1} , cm^3 and cal/mol .

No.	Reactions	A	n	E _a	References
R1	EV = C ₄ H ₉ COOH + C ₂ H ₄				[18] ^a
	PLOG / 0.0000001	5.248E+02	−3.590	1.795E+04/	
	PLOG / 0.0001316	1.380E+78	−20.180	7.260E+04/	
	PLOG / 0.0013158	1.585E+85	−21.730	8.170E+04/	
	PLOG / 0.0131579	2.754E+85	−21.330	8.724E+04/	
	PLOG / 0.1315789	1.202E+85	−20.820	9.211E+04/	
	PLOG / 1	6.166E+41	−8.200	6.715E+04/	
	PLOG / 1.1842105	6.166E+39	−7.620	6.603E+04/	
	PLOG / 10,000	2.089E+13	0.000	4.954E+04/	
R2	EV + H = EV2J + H ₂	9.120E+09	1.220	4.770E+03	[46] ^b
R3	EV + H = EV3J + H ₂	1.430E+10	1.200	6.050E+03	[46] ^b
R4	EV + H = EV4J + H ₂	1.430E+10	1.200	6.050E+03	[46] ^b
R5	EV + H = EV5J + H ₂	6.070E+10	1.180	8.600E+03	[46] ^b
R6	EV + H = EVEJ + H ₂	1.440E+10	1.250	5.970E+03	[46] ^b
R7	EV + H = EVMJ + H ₂	4.830E+10	1.220	9.860E+03	[46] ^b
R8	EV + CH ₃ = EV2J + CH ₄	2.075E+04	2.260	7.287E+03	[9]
R9	EV + CH ₃ = EV3J + CH ₄	2.705E+04	2.260	7.287E+03	[9]
R10	EV + CH ₃ = EV4J + CH ₄	2.705E+04	2.260	7.287E+03	[9]
R11	EV + CH ₃ = EV5J + CH ₄	4.520E-01	3.650	7.154E+03	[9]
R12	EV + CH ₃ = EVEJ + CH ₄	1.990E+01	3.370	7.635E+03	[9]
R13	EV + CH ₃ = EVMJ + CH ₄	4.520E-01	3.650	7.154E+03	[9]
R14	EV + OH = EV2J + H ₂ O	4.660E+02	3.040	−2.160E+03	[43] ^c
R15	EV + OH = EV3J + H ₂ O	7.760E-01	3.760	−4.890E+03	[43] ^c
R16	EV + OH = EV4J + H ₂ O	7.760E-01	3.760	−4.890E+03	[43] ^c
R17	EV + OH = EV5J + H ₂ O	4.380E+04	2.610	7.500E+02	[43] ^c
R18	EV + OH = EVEJ + H ₂ O	2.800E+00	3.670	−4.131E+03	[43] ^c
R19	EV + OH = EVMJ + H ₂ O	2.589E+00	3.730	−3.790E+02	[43] ^c
R20	EV + O = EV2J + OH	4.770E+04	2.710	2.106E+03	[9]
R21	EV + O = EV3J + OH	4.770E+04	2.710	2.106E+03	[9]
R22	EV + O = EV4J + OH	4.770E+04	2.710	2.106E+03	[9]
R23	EV + O = EV5J + OH	9.650E+04	2.680	3.716E+03	[9]
R24	EV + O = EVEJ + OH	1.450E+05	2.410	8.760E+02	[9]
R25	EV + O = EVMJ + OH	9.650E+04	2.680	3.716E+03	[9]
R26	EV + HO ₂ = EV2J + H ₂ O ₂	1.080E-03	4.410	1.140E+04	[44] ^c
R27	EV + HO ₂ = EV3J + H ₂ O ₂	9.140E-04	4.520	1.158E+04	[44] ^c
R28	EV + HO ₂ = EV4J + H ₂ O ₂	1.120E+00	3.620	1.361E+04	[44] ^c
R29	EV + HO ₂ = EV5J + H ₂ O ₂	2.690E+02	3.100	1.714E+04	[44] ^c
R30	EV + HO ₂ = EVEJ + H ₂ O ₂	3.780E-05	4.900	1.069E+04	[44] ^c
R31	EV + HO ₂ = EVMJ + H ₂ O ₂	4.110E-01	3.980	1.873E+04	[44] ^c

Note.

^a Referred to similar reaction of ethyl levulinate.^b referred to similar reaction of ethyl butanoate.^c taken from site-specific rate rules for H-abstraction reactions of esters by OH and HO₂.

estimated following the analogy strategy. Pressure-dependent rate constants of the C–C bond dissociation reactions in the valeric acid group were referred to analogy reactions of methyl valerate [50]. The rest C–C and C–O bond dissociation reactions in the acyl group and ethyl group were referred to similar reactions of ethyl propionate [15]. HPL rate constants of these reactions are compared in Fig. 2. It is observed that the rate constant of the C(=O)–O bond dissociation reaction is the lowest one among all bond dissociation reactions, which is mainly due to its highest bond dissociation energy [51,52].

3.2. H-abstraction reactions of ethyl valerate

H-abstraction reactions of the fuel by reactive species such as H, OH and HO₂ radicals play an important role in combustion. Mendes et al. [43,44] performed theoretical investigations on H-abstraction reactions of a series of esters by OH and HO₂ radicals, providing a rate rule at different carbon positions. Thus, the rate constants based on their rate rule were adopted in the present model. No investigation has been conducted for H-abstraction reactions of ethyl valerate by H. In this work, the rate constants were referred to similar reactions of ethyl butanoate calculated by Wang et al. [46], while same rate constants were used for H-abstraction reactions at β and γ carbon sites as both sites are secondary carbon site. For H-abstraction reactions of ethyl valerate by

O, O₂ and methyl (CH₃), the rate constants were taken from the Dayma model [9], which were mainly referred to similar reactions of methyl esters.

3.3. Decomposition reactions of fuel radicals

For the six radicals of ethyl valerate formed from H-abstraction reactions, two reaction classes were incorporated in the present model, i.e. β -scission reactions and isomerization reactions involving 5-, 6- and 7-membered transition states. There are few investigations on the rate constants of decomposition reactions of ethyl ester radicals and the competition between the two reaction classes in literature, so the analogy method to methyl esters was adopted. In the present model, the β -C–C and β -C–O scission reactions were mainly referred to those of methyl butanoate radicals calculated by Huynh et al. [53] and methyl decanoate radicals calculated by Meng et al. [54]. Ethyl valerate radicals can also undergo the β -C–H scission reactions producing ethyl pentenoate with double bond at $C_{\alpha}=C_{\beta}$, $C_{\beta}=C_{\gamma}$ or $C_{\gamma}=C_{\delta}$ and vinyl valerate. The rate constants of these reactions were taken from the Dayma model [9]. Isomerization reactions between six fuel radicals were also incorporated in the present model, while the rate constants were referred to the calculation work of methyl butanoate radicals [53] or taken from the Dayma model [9].

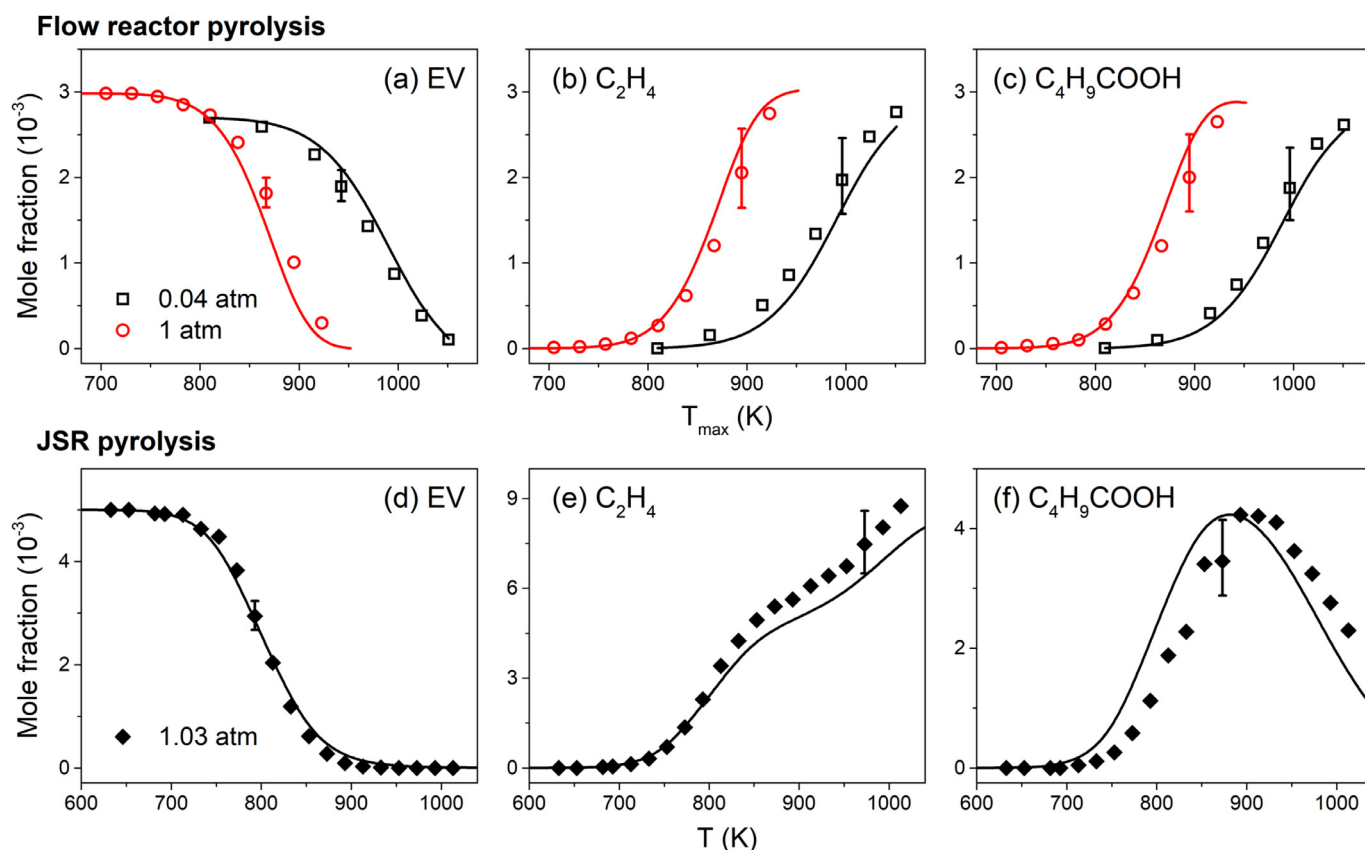


Fig. 3. Measured (symbols) and simulated (lines) mole fraction profiles of ethyl valerate, ethylene and valeric acid in (a–c) flow reactor pyrolysis of ethyl valerate at 0.04 and 1 atm and (d–f) JSR pyrolysis of ethyl valerate at 1.03 atm.

3.4. Sub-mechanism of valeric acid

Recently, Namysl et al. [42] proposed an oxidation model of butyric and valeric acid to simulate the JSR oxidation experiments. Rate constants of reactions in the sub-mechanisms of these acids were mainly estimated using the analogy strategy or taken from recent theoretical calculations. In the present model, the sub-mechanism of valeric acid were adopted from their work. For pentenoic acid isomers, the H-abstraction reactions and subsequent decomposition reactions of pentenoic acid radicals were lumped in the present model following the strategy of Namysl et al. [42]. The present model was also validated against the JSR oxidation data of valeric acid reported by Namysl et al. [42]. Reasonable performance of the present model in predicting their data can be observed from Figs. S2 and S3 in the *Supplementary Materials*.

4. Results and discussion

4.1. Flow reactor and jet-stirred reactor pyrolysis

A series of pyrolysis products were observed in the flow reactor and JSR pyrolysis experiments. The measured mole fraction profiles of these species as a function of temperature were compared with the simulated results of the present model. For the simulation of flow reactor pyrolysis, the Plug Flow Reactor module in the Chemkin-Pro software [55] was used with the measured temperature profiles as input parameters. For the simulation of JSR pyrolysis, the Perfectly Stirred Reactor module in the Chemkin-Pro software [55] was used with Fixed Gas Temperature as the problem type and reactor volume and inlet flow rate as input parameters.

4.1.1. Decomposition of ethyl valerate

The measured and simulated mole fraction profiles of ethyl valerate, ethylene and valeric acid in both flow reactor and JSR pyrolysis are shown in Fig. 3. It can be observed that the present model can reasonably predict the decomposition of ethyl valerate and formation of ethylene and valeric acid. Modeling analysis, including rate of production (ROP) analysis and sensitivity analysis, was performed to explore the chemistry in ethyl valerate pyrolysis. Over the whole temperature ranges, the ROP analysis suggests that ethyl valerate is almost exclusively consumed through R1, while the sensitivity analysis also highlights the dominant role of R1 to the consumption of ethyl valerate. Fig. 4(a) shows that among the decomposition reactions of ethyl valerate, R1 is the only one with observable sensitivity coefficients over the whole temperature range in the sensitivity analysis of ethyl valerate at 0.04 atm. This is dramatically different from the situation in the pyrolysis of methyl esters [56–58] where the intramolecular elimination reactions are only minor decomposition pathways of the fuels under similar conditions due to the absence of β' -C-H bond to undergo the six-membered intramolecular elimination reaction. Thus, the pyrolysis data of ethyl valerate can provide crucial constraints to the rate constant of R1.

Based on above discussion, it can be concluded that the good performance of the present model in reproducing the measured mole fraction profiles of ethyl valerate and the two major products results from the incorporation of the updated rate constants of R1. In comparison, performance of the Dayma model [9] and the Dmitriev model [11] on predicting the flow reactor pyrolysis data at 0.04 and 1 atm are shown in Fig. S4 in the *Supplementary Materials*. Their estimated rate constant of R1 is about half as the

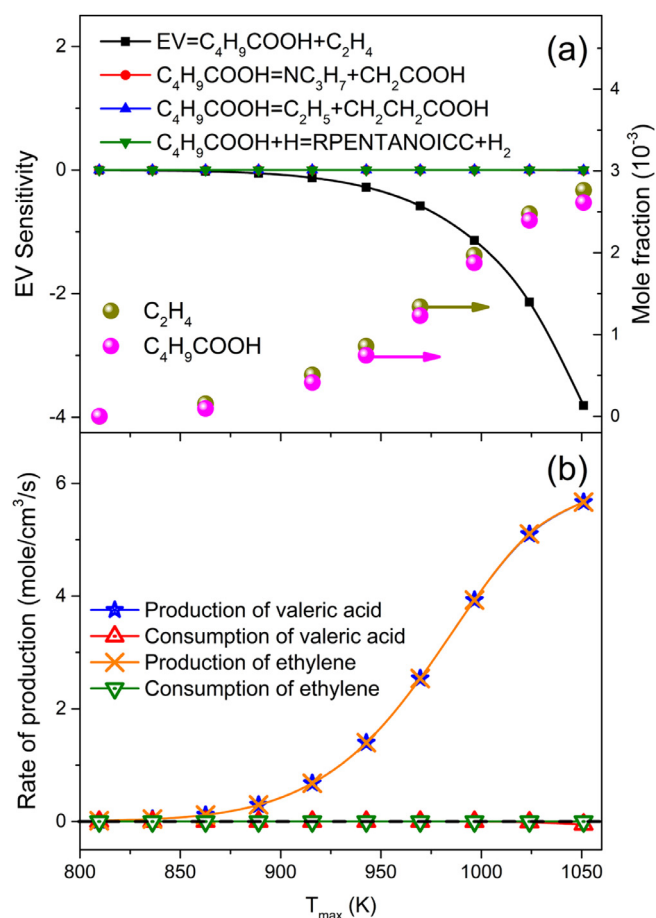


Fig. 4. (a) Sensitivity analysis of ethyl valerate and measured mole fraction profiles of valeric acid and ethylene and (b) ROP analysis for production and consumption of valeric acid and ethylene in the flow reactor pyrolysis of ethyl valerate at 0.04 atm.

one used in the present model. Consequently Fig. S4 shows that their model under-predicts the pyrolysis reactivity of ethyl valerate at both 0.04 and 1 atm.

Under the investigated conditions (gray zone highlighted in Fig. 2), at least 3 orders of magnitude differences are observed for the rate constants between R1 and the C–C and C–O bond dissociation reactions, which explains the exclusive contribution of R1 to ethyl valerate consumption. This also differs with the situation of methyl esters [56] where the bond dissociation dominates over the intramolecular elimination under similar conditions. Consequently, ethylene and valeric acid become the two dominant primary decomposition products of ethyl valerate. As can be observed from Fig. 3, their initial formation temperatures are almost the same as the initial decomposition temperatures of ethyl valerate. Simultaneous formation and close concentrations of the two products can also be observed from Fig. 4(a). The ROP analysis results in Fig. 4(b) reveal that both ethylene and valeric acid are scarcely consumed, thus their concentrations keep accumulated until the fuel is almost totally consumed.

4.1.2. Consumption of valeric acid

In the flow reactor pyrolysis, weak signals of methyl, allene (AC_3H_4), propyne (PC_3H_4), propene (C_3H_6), 1-butene (C_4H_8-1), 2-butene (C_4H_8-2) and *n*-butane (C_4H_{10}) were also detected at 0.04 atm. However, their concentrations could hardly be evaluated with high precision due to the low signal-to-noise ratios. The ROP analysis reveals that over the measured temperature regions in the flow reactor pyrolysis, methyl, propene, 1-butene and *n*-butane are

mainly produced by subsequent reactions of valeric acid with extraordinarily small reaction fluxes, which exists until the depletion temperatures of ethyl valerate. Two reasons are responsible for this situation. First, the decomposition of ethyl valerate is dominated by R1 producing ethylene and valeric acid under the investigated conditions, as mentioned above. Secondly, though valeric acid is the less stable one in the two products of R1, its unimolecular decomposition is still not prone to happen before the depletion temperatures of ethyl valerate. As a result, the chain initiation process is inhibited in the pyrolysis of ethyl valerate over a large temperature range after the initial decomposition of the fuel, which is also a feature compared with the pyrolysis of methyl esters where the chain initiation process occurred almost simultaneously with the initial decomposition of the fuels [56].

In the JSR pyrolysis, the measurements were performed over a wide temperature region. The maximum measured temperature is more than 100 K higher than the depletion temperature of ethyl valerate. As a result, the decomposition reaction of valeric acid occurred and a series of secondary decomposition products were observed. In particular, the ROP analysis shows that C–C bond dissociation reactions of valeric acid enhance the chain initiation process and greatly enrich the radical pool. The produced radicals can attack valeric acid to promote the consumption, while ethylene can hardly be intensively consumed under the investigated conditions. Fig. 5 displays the observed decomposition products of valeric acid in the JSR pyrolysis, showing that the present model can reasonably simulate the formation of these products.

Fig. 6 draws the consumption pathways of valeric acid at 1000 K and 1.03 atm in the JSR pyrolysis based on the ROP analysis, while the carbon sites of valeric acid are defined following the rule of ethyl valerate in Fig. 1. Both C–C bond dissociation reactions and H-abstraction reactions contribute to the consumption of valeric acid. According to the calculations by Namysl et al. [42], the dissociation energy of $C_{\alpha}-C_{\beta}$ bond is at least 7 kcal/mol lower than those of the other bonds in valeric acid. Consequently, the $C_{\alpha}-C_{\beta}$ bond dissociation reaction of valeric acid (R32) has the largest contribution to the consumption of valeric acid (27%), forming *n*-propyl (NC_3H_7) and a resonantly stabilized radical CH_2COOH . Besides unimolecular decomposition reactions, valeric acid can also be consumed through H-abstraction reactions to produce four radicals, i.e. $CH_3(CH_2)_2\cdot CHC(=O)OH$ (RPENTANOICA), $CH_3CH_2\cdot CHCH_2C(=O)OH$ (RPENTANOICB), $CH_3\cdot CH(CH_2)_2C(=O)OH$ (RPENTANOICC) and $\cdot CH_2(CH_2)_3C(=O)OH$ (RPENTANOICD). Among the reactive species initiating H-abstraction, methyl is the most important one and the H-abstraction reactions by methyl contribute 35% to the consumption of valeric acid at 1000 K, while those by H and OH contribute 23% and 12%, respectively. Fig. 6 also shows that the H-abstraction reactions at α , β and γ carbon sites have very close contributions, while the H-abstraction reactions at δ carbon site only contribute half of them. Besides the H-abstraction reactions at carbon sites, the H-abstraction reactions at the sole oxygen site contribute less than 1% due to the much higher dissociation energy of O–H bond than those of C–H bonds (at least 5.5 kcal/mol) [42], thus these reactions are not shown in Fig. 6.



For the two products of R32, the decomposition of *n*-propyl is dominated by the β -C–C scission reaction to form ethylene and methyl (R33). Two unimolecular decomposition reactions of CH_2COOH , one producing methyl and CO_2 (R34) and the other one producing ketene (CH_2CO) and OH (R35), were suggested as the key pathways by Cavallotti et al. [59], while their calculated rate constants were adopted in the present model. The branching ratio of R34 can be up to 89% at 1000 K and 1 atm. Thus, the two products of R32 can both be consumed to form methyl through

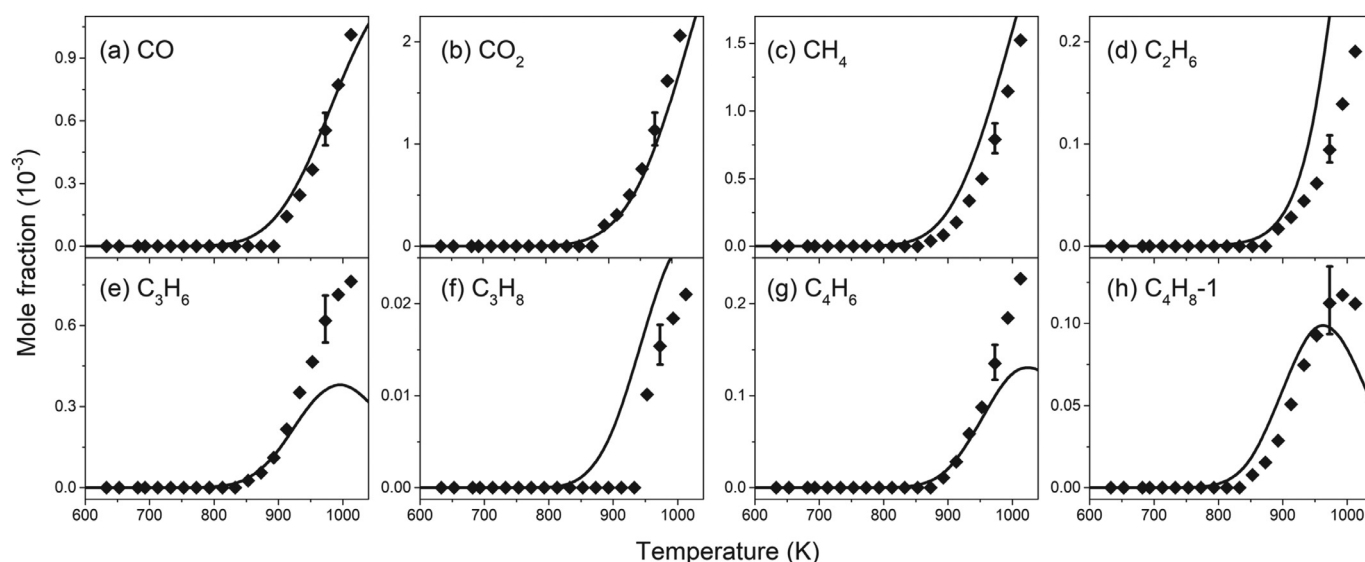


Fig. 5. Measured (symbols) and simulated (lines) mole fraction profiles of (a) CO, (b) CO₂, (c) methane, (d) ethane, (e) propene, (f) propane, (g) 1,3-butadiene and (h) 1-butene in the JSR pyrolysis of ethyl valerate at 1.03 atm.

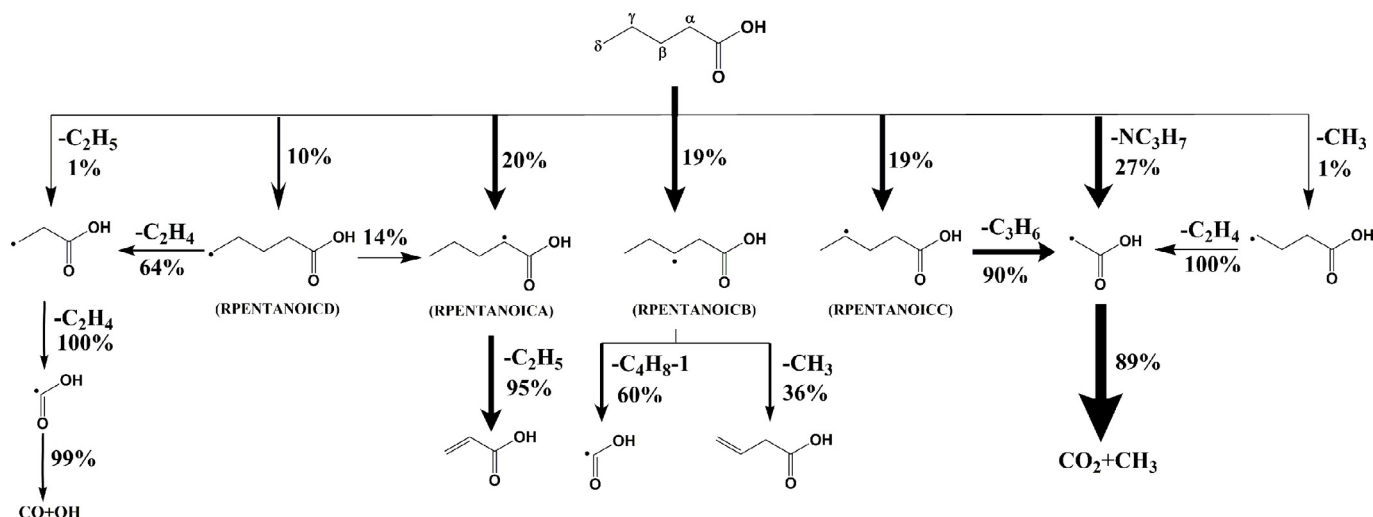


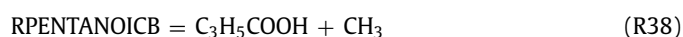
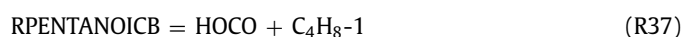
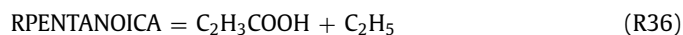
Fig. 6. Main reaction network of valeric acid in the JSR pyrolysis of ethyl valerate at 1000 K and 1.03 atm. The percentage on each arrow denotes the contribution of this pathway to the consumption of corresponding reactant.

R33 and R34, respectively. The abundant production of methyl explains the significant role of H-abstraction reactions by methyl in the consumption of valeric acid at 1000 K. In addition, R33 also enhances the formation of ethylene and R34 contributes 90% to CO₂ formation.



The valeric acid radicals formed from the H-abstraction reactions of valeric acid can undergo β -scission reactions and isomerization reactions. Products with C=C double bond can be produced in the subsequent decomposition of valeric acid radicals. For RPENTANOICA and RPENTANOICC, the dominant decomposition pathways are β -C-C scission reactions producing acrylic acid (C₂H₃COOH) + ethyl (C₂H₅) (R36) and CH₂COOH + propene (R39), respectively. For RPENTANOICB, two β -C-C scission reactions contribute to its decomposition producing HOCO and 1-butene (R37)

and 3-butenic acid (C₃H₅COOH) + methyl (R38) with a branching ratio of around 1:2. R37 and R39 also contribute approximately 80% and 68% to the formation of 1-butene and propene, respectively. For RPENTANOICD, it has two kinds of main consumption reactions, i.e. the β -C-C scission reaction producing CH₂CH₂COOH and ethylene (R40) and 1,4-H transfer reaction to form RPENTANOICA (R41). The former one is the dominant consumption reaction of the RPENTANOICD while the latter contributes about 14% to its consumption. In addition, the four radicals of valeric acid can also undergo the β -C-H scission reactions producing corresponding pentenoic acid isomers and H, which are minor consumption pathways compared with β -C-C scission reactions and are not shown in Fig. 6. In the present model, all the produced pentenoic acid isomers were lumped as 4-pentenoic acid and its sub-mechanism was taken from Namysl et al. [42].



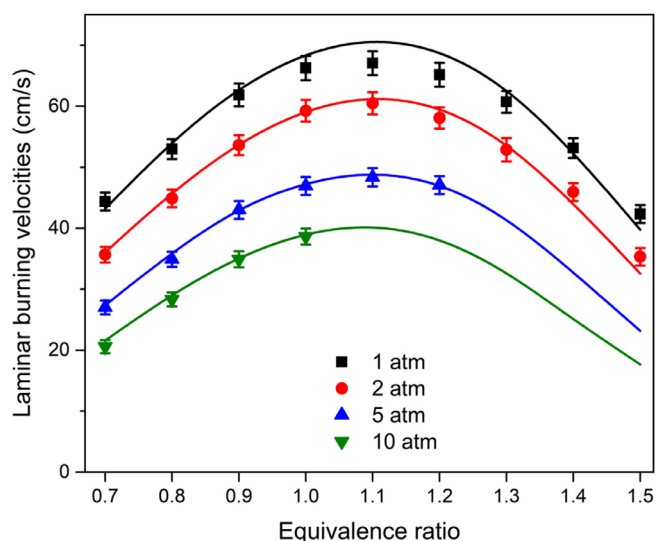


Fig. 7. Measured (symbols) and simulated (lines) LBVs of ethyl valerate/air mixtures at $T_u = 443$ K and $P_u = 1$ –10 atm.

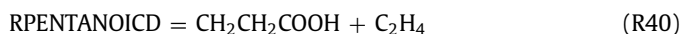
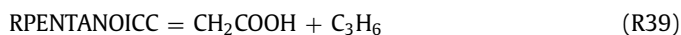


Fig. 3(e,f) shows that valeric acid has a peak mole fraction over the wide temperature region in the JSR pyrolysis, while the mole fraction of ethylene keeps growing as temperature increases. As discussed above, when temperature is lower than 900 K, the intensive growing of mole fraction profiles of both valeric acid and ethylene is due to the large reaction flux of R1 and negligible consumption of the two products. When temperature is higher than 900 K, the decomposition of valeric acid becomes accelerated and subsequent reactions lead to further formation of ethylene. At 1000 K, the ROP analysis indicates that the contribution of R1 to the formation of ethylene decreases to 66%, while the decomposition reactions of valeric acid contribute the rest together. Meanwhile, there is still only slight consumption of ethylene at 1000 K. As a result, valeric acid has a peak-shape mole fraction profile and ethylene has an ascending mole fraction profile.

4.2. Laminar burning velocity

In this work, LBVs of ethyl valerate were measured at $T_u = 443$ K, $P_u = 1, 2, 5$ and 10 atm and various equivalence ratios. The LBVs were simulated using the Premixed Laminar Flame Speed Calculation module of the Chemkin-Pro software [55] with the consideration of Soret effect. Mixture-averaged transport and sufficient refinement over a 10 cm domain with a maximum grid point of 1000 (GRAD 0.02, CURV 0.02) were used during simulation. Fig. 7 compares the measured and simulated LBVs of ethyl valerate. It is observed that the present model can reasonably predict the measured data under the investigated conditions. At 1, 2 and 5 atm, the LBVs of ethyl valerate peak at around $\phi = 1.1$. At 5 and 10 atm, available measured LBVs are limited to $\phi = 1.2$ and 1.0, respectively, due to the growing influence of cellular instabilities under rich and high-pressure conditions. As expected, the LBV results show a descending trend as P_u increases.

To reveal the critical reactions in the laminar flame propagation of ethyl valerate, the ROP analysis and sensitivity analysis under lean ($\phi = 0.7$) and rich ($\phi = 1.5$) conditions at $P_u = 1$ and 10 atm were performed. The ROP analysis results shown in Fig. 8 indicate

that even under flame conditions with a more abundant radical pool than pyrolysis conditions, R1 still has noticeable contributions to the consumption of ethyl valerate, varying from 14% at $P_u = 1$ atm and $\phi = 0.7$ to 49% at $P_u = 10$ atm and $\phi = 1.5$. The rest of ethyl valerate is mainly consumed via H-abstraction reactions, which is the main difference between the flame conditions and the pyrolysis conditions. Among the reactive species initiating H-abstraction, H, OH and O radicals are dominant ones, while methyl is a minor one.

As a product of R1, valeric acid is mainly consumed by H-abstractions under flame conditions. The contribution can reach up to 91% under lean conditions at $P_u = 1$ atm. Similar to the situation under pyrolysis conditions, the following consumption pathways of valeric acid radicals are also β -scission reactions leading to ethylene, CO_2 and CO. For the H-abstraction reactions, the products are six radicals of ethyl valerate with the radical site at α , β , γ , δ , α' or β' carbon site, referred as EV2J, EV3J, EV4J, EV5J, EVEJ or EVMJ radical, respectively. These fuel radicals can undergo β -scission reactions and isomerization reactions. As shown in Fig. 8, β -C-C and β -C-O scission reactions producing a smaller radical and a molecule with C=C or C=O double bond dominate the consumption of EV4J, EV5J, EVEJ and EVMJ radicals. For EV2J and EV3J radicals, besides the important β -C-C scission reactions, β -C-H scission reactions yielding ethyl 2-pentenoate (EV2D) and ethyl 3-pentenoate (EV3D) and isomerization reactions involving 5-, 6- and 7-membered transition states also contribute to their consumptions. At different pressures and equivalence ratios, there is little difference in the contributions among these reactions. Subsequent decomposition pathways for the products of fuel radicals are also shown in Fig. 8, which reveals that ethylene, acetaldehyde (CH_3CHO), ketene, CO_2 and CO are the major products in the decomposition reaction sequences of fuel radicals.

Based on the above analysis, ethylene can be abundantly produced in ethyl valerate flames, which was also verified by the experimental observations in previous low-pressure premixed flame study of ethyl valerate [10]. In particular, R1 and β -scission reactions of EV5J, EVMJ and successive products of other ethyl valerate radicals, such as ethyl, *n*-propyl, *n*-butyl (PC_4H_9) and RPENTANOICD, all contribute to ethylene formation. The ROP analysis shows that these β -scission reactions have total contributions of 72% at $P_u = 1$ atm and $\phi = 0.7$ and 66% at $P_u = 10$ atm and $\phi = 1.5$. Ethylene is mainly consumed via the H-abstraction reactions by H and OH producing vinyl (C_2H_3) (R42 and R43) and the addition-elimination reactions with O (R44–R46) under all investigated conditions.



Fig. 9 shows the results of sensitivity analysis for LBVs of ethyl valerate under lean and rich conditions at $P_u = 1$ and 10 atm. It can be found that only C_0 – C_2 related reactions are involved in the most sensitive reactions, while reactions involving ethyl valerate and valeric acid are much less sensitive. Under rich conditions, the most important chain-branching reaction $\text{H} + \text{O}_2 = \text{O} + \text{OH}$ is the prevailing sensitive reaction, while under lean conditions, the most important chain-propagation reaction $\text{CO} + \text{OH} = \text{CO}_2 + \text{H}$ has comparative sensitivity coefficients to $\text{H} + \text{O}_2 = \text{O} + \text{OH}$. Among the reactions with negative sensitivity coefficients, the chain-inhibition reaction $\text{H} + \text{O}_2 (+\text{M}) = \text{HO}_2$

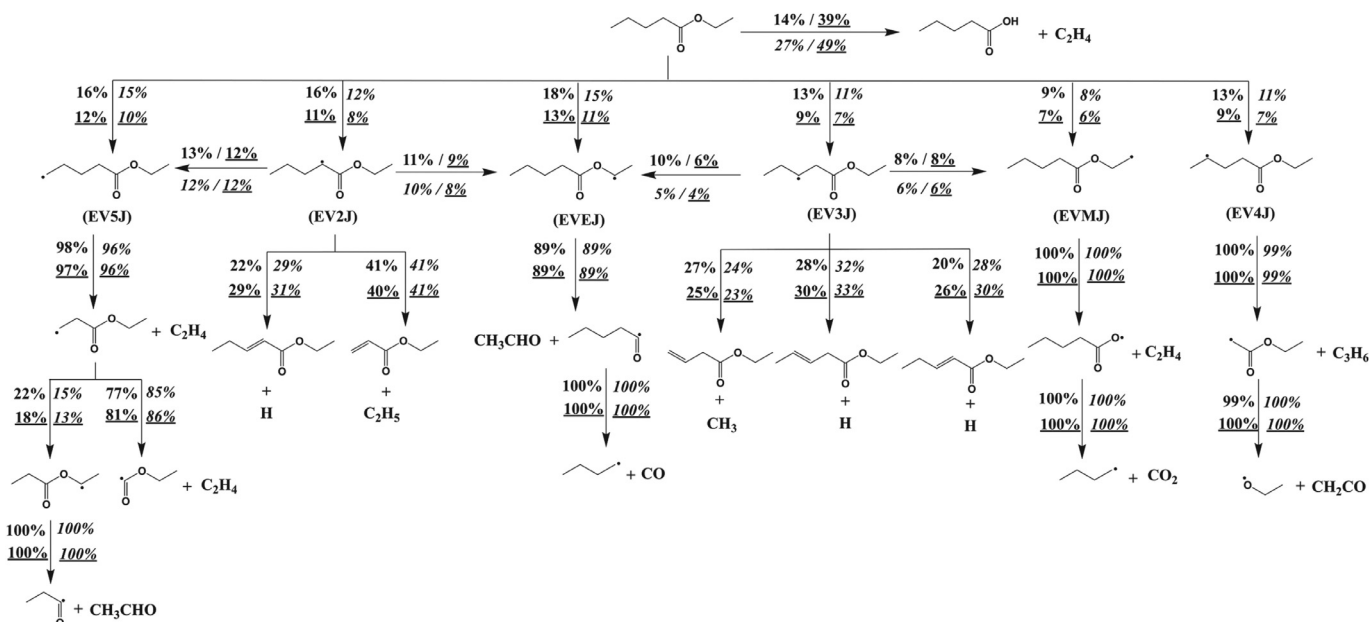


Fig. 8. Main reaction network of ethyl valerate decomposition in laminar flame propagation. The percentage on each arrow denotes the contribution of this pathway to the consumption of corresponding reactant at $P_u = 1$ atm and 10 atm (underlined) and $\phi = 0.7$ and 1.5 (italic).

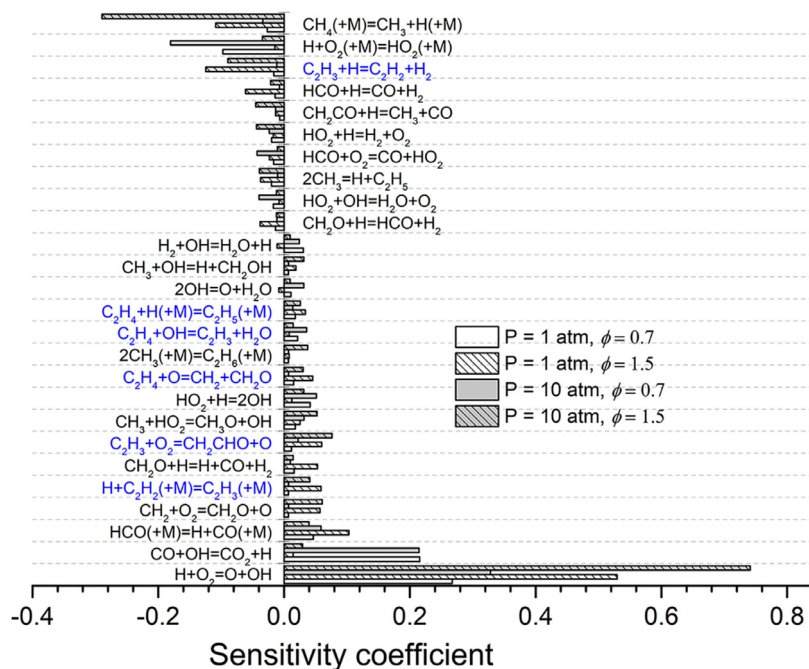


Fig. 9. Sensitivity analysis of LBVs of ethyl valerate at $T_u = 443$ K and $P_u = 1$ –10 atm. Ethylene- and vinyl-involved reactions are highlighted in blue. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

(+M) has the greatest negative effects under lean conditions due to the abundant concentration of O_2 , while chain-termination reactions, i.e. $C_2H_3 + H = C_2H_2 + H_2$ (R47) and the reverse reaction of $CH_4 (+M) = CH_3 + H (+M)$, have the greatest negative effects under rich conditions. The crucial role of R47, especially under rich conditions at 1 atm, can be explained by the abundant production of ethylene in ethyl valerate flames. Similar importance of this reaction was also found in the laminar flame propagation of ethylene [60]. Besides R47, vinyl can also react with O_2 to generate CH_2CHO and O (R48), which is a chain-branching reaction and has positive sensitivity coefficients. Furthermore, a series of other reactions involving ethylene and vinyl with positive sensitivity coefficients

can also be observed in Fig. 9. Among them, R43 is more sensitive under lean conditions, while other reactions are more sensitive under rich conditions. The positive sensitivity coefficients of R43 and β -C-H scission reactions of ethyl and vinyl result from the direct or indirect H yield, while the positive sensitivity coefficients of R45 are related to the HCO formation from further reactions of both products, i.e. triplet methylene (CH_2) and formaldehyde (CH_2O).



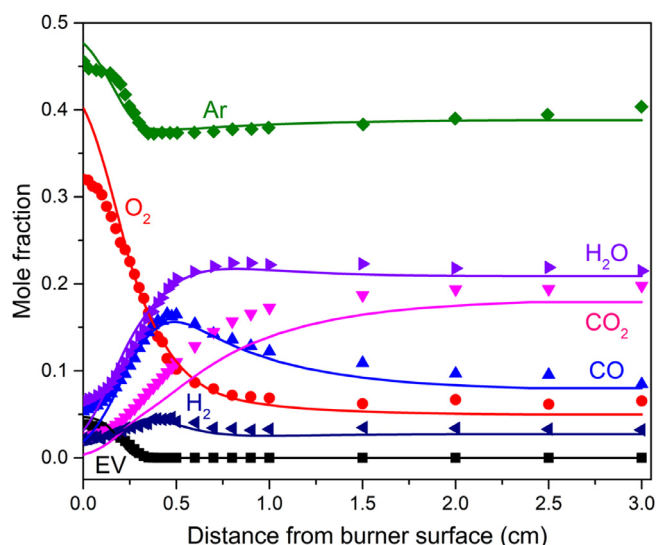


Fig. 10. Comparison between measured mole fraction profiles of major species (ethyl valerate, O_2 , Ar, H_2 , H_2O , CO and CO_2) in the laminar premixed flame of ethyl valerate at 0.026 atm and $\phi = 1.0$ [10] (symbols) and simulated results of the present model (lines).

4.3. Extended validation of the present model against ethyl valerate data in literature

4.3.1. Validation against laminar premixed flame data

Several literature efforts [10–13] have also been dedicated to investigate the chemical structures of low-pressure and atmospheric-pressure premixed flames of ethyl valerate. Knyazkov et al. [10] measured the stoichiometric premixed flame of ethyl valerate at 0.026 atm using SVUV-PIMS. Dmitriev et al. [11] investigated the

stoichiometric premixed flame of ethyl valerate at 0.066 atm and 1 atm using GC and MBMS, respectively. Katshiatshia et al. [12,13] measured the ethyl valerate premixed laminar flames at 0.054 atm and various equivalence ratios ($\phi = 0.81, 0.95$ and 1.31) using GC. These measured speciation profiles of radicals and stable species present useful validation targets for the present model. The simulation work was performed using the Premixed Laminar Burner-Stabilized Flame module in the Chemkin-Pro software [55] with the measured temperature profiles as input parameters. The experimental and simulated mole fraction profiles of major flame species (ethyl valerate, O_2 , Ar, H_2 , H_2O , CO and CO_2) and intermediates are shown in Figs. 10–11 and Figs. S5–S9 in the *Supplementary Materials*. It can be observed that the present model can reasonably predict most of the species, while generally under-predicts the results of C_3 hydrocarbons, such as propene, allyl (AC_3H_5), allene (AC_3H_4) and propyne (PC_3H_4) under low-pressure conditions. The under-predictions of C_3 hydrocarbons were also observed by Knyazkov et al. [10] using the Dayma model [9]. Under the low-pressure flame conditions, ROP analysis reveals that ethyl valerate is mainly consumed by H-abstraction reactions by H, OH and O radicals, while the contributions of R1 is around 13% at 0.026 atm and $\phi = 1.0$. As the equivalence ratio increases, the contribution of R1 increases. Fate of these produced fuel radicals are β -scission reactions and isomerization reactions, which are illustrated in Fig. 8. As seen from Fig. 11, ethylene and acetaldehyde are abundantly produced in the ethyl valerate flames, which is consistent with the above analysis for LBVs.

4.3.2. Validation against LBV measurements at various initial temperatures and pressures

Besides the present LBV measurements, there are several previous work measuring the LBVs of ethyl valerate at various initial temperatures and pressures. Katshiatshia et al. [13] measured the LBVs of ethyl valerate at $P_u = 1$ atm and $T_u = 318, 328$ and 338 K

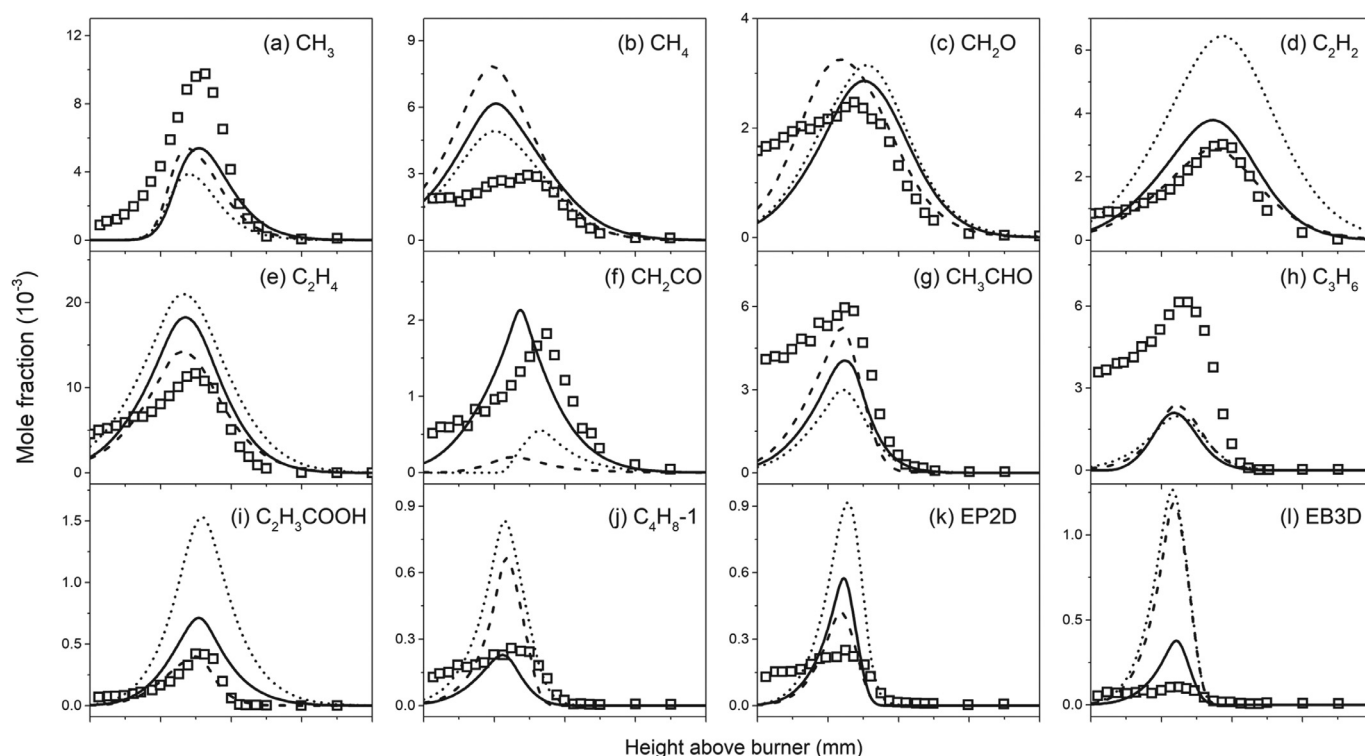


Fig. 11. Comparison between the measured mole fraction profiles of (a) methyl, (b) methane, (c) formaldehyde, (d) acetylene, (e) ethylene, (f) ketene, (g) acetaldehyde, (h) propene, (i) acroleic acid, (j) 1-butene, (k) ethyl-2-propenoate, (l) ethyl-3-butenate in the laminar premixed flame of ethyl valerate at 0.026 atm and $\phi = 1.0$ [10] (symbols) and simulated results of the present model (solid lines), the Dayma model [9] (dash lines) and the Dmitriev model [11] (dot lines).

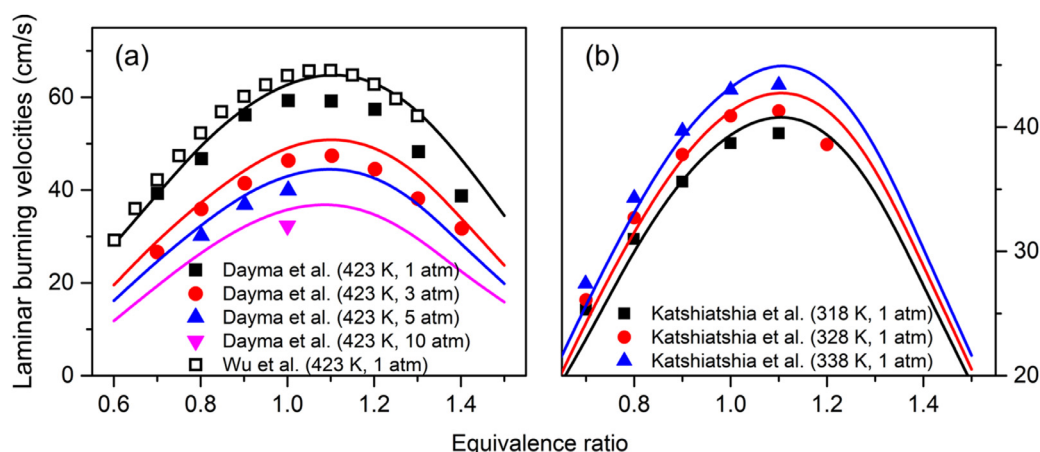


Fig. 12. Comparison between measured LBVs of ethyl valerate (symbols) at various initial temperatures and pressures [9,13,14] and simulated results of the present model (lines).

using the heat flux burner method, while a higher initial temperature at 423 K was measured by Wu et al. [14] at $P_u = 1$ atm using the Bunsen flame method. Dayma et al. [9] measured the LBVs of ethyl valerate at $T_u = 423$ K and $P_u = 1, 3, 5$ and 10 atm using outwardly propagating spherical flame method. The simulations were also conducted using the Premixed Laminar Flame Speed Calculation module of the Chemkin-Pro software [55] and the comparison between the measured and simulated results are shown in Fig. 12. It is seen that there exists discrepancies between the measurements of Wu et al. [14] and Dayma et al. [9] under stoichiometric and rich conditions at 1 atm, while the present model is in more accordance with the data of Wu et al. [14] under these conditions. In general, the present model exhibits reasonable performance in reproducing the LBVs of ethyl valerate at various T_u and P_u .

5. Conclusions

Pyrolysis and laminar flame propagation of ethyl valerate were investigated in this work under various conditions by the combination of SVUV-PIMS, GC and high-speed Schlieren techniques. A high-temperature kinetic model of ethyl valerate was developed to reproduce and interpret the experimental observations in above pyrolysis speciation and LBV measurements with the focus on key intermediates and reactions. Major conclusions are drawn as follows.

- (1) Under the pyrolysis conditions, the intramolecular elimination reaction of ethyl valerate producing valeric acid and ethylene exclusively contributes to the fuel consumptions due to its low energy barrier. The highly sensitive reaction system is helpful to constrain the pressure-dependent rate constant of this reaction. Under the higher temperature conditions in the JSR pyrolysis experiment, the decomposition of valeric acid becomes important after the depletion of fuel. In particular, its C–C bond dissociation reactions open the chain initiation process and greatly enrich the radical pool.
- (2) Under the flame conditions, significant contributions of the intramolecular elimination reaction to the consumption of ethyl valerate can also be observed, especially under rich conditions. C_0 – C_1 reactions have the highest sensitivity coefficients to the flame propagation, while ethylene- and vinyl-involved reactions are also among sensitive reactions due to the abundant production of ethylene.
- (3) Besides the experimental data obtained in this work, the present model was also validated against the available literature data of ethyl valerate and valeric acid, including the JSR

oxidation at near-atmospheric pressure, low-pressure premixed flames at various equivalence ratios and LBVs at different T_u and P_u . In general, the present model can present reasonable simulation on both the new experimental data in this work and the data in literature.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.combustflame.2020.12.051](https://doi.org/10.1016/j.combustflame.2020.12.051).

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